

MANDEEP SIWACH

Data Engineer | ETL & Pipeline Development | GCP | SQL & Python

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PROFESSIONAL SUMMARY

Data Engineer with hands-on experience building end-to-end data pipelines, production-grade tracking portals, and SAP-integrated automation systems. Proficient in Python, SQL, PostgreSQL, ETL workflows, and Power BI. Currently expanding expertise in GCP (BigQuery, Dataflow, Pub/Sub) and preparing for the GCP Professional Data Engineer certification. Strong background in process automation, data quality enforcement, and supply-chain analytics.

SKILLS

Languages	Python, SQL, Java
Databases	PostgreSQL, MySQL, BigQuery
ETL & Pipelines	Apache Spark, Apache Airflow, Dataflow, Pub/Sub
Cloud (GCP)	BigQuery, Cloud Run, Cloud Storage, Dataproc, Pub/Sub
BI & Viz	Power BI, Excel, Matplotlib, Seaborn
Frameworks	FastAPI, Spring Boot, Pandas, NumPy, Scikit-Learn
Infra / OS	Linux VM, Internal Server, Git
Soft Skills	Problem-Solving, Attention to Detail, Process Ownership

WORK EXPERIENCE

Data & Systems Engineer · VVDN Technologies · Manesar

Jun 2024 – Present

- Designed and built an end-to-end project tracking portal covering the full lifecycle (procurement → production → dispatch) with mandatory field validation to enforce process compliance and catch errors at source.
- Deployed and maintained the portal on a Linux-based internal server VM — managed access control, production uptime, and server configuration end-to-end without external support.
- Integrated SAP APIs to auto-validate invoice vs. PO data before dispatch — eliminated mismatches and prevented financial discrepancies.
- Implemented duplicate Purchase Request detection logic to flag cases where multiple PRs were raised for the same PO in SAP, reducing redundant procurement entries.
- Designed stage-wise bottleneck tracking dashboard identifying which production stage and team caused the most delays — enabling targeted process improvements.
- Built auto-checkpoints at every workflow step to enforce data quality and sign-off conditions before progression.
- Developed QR-code-based Bins Management system — each bin tagged with QR linked to material details for real-time inventory tracking.
- Leveraged AI-assisted development (prompt engineering) to accelerate portal backend delivery and automate complex workflow logic.
- Automated PPC workflows from raw data ingestion to Power BI dashboard, eliminating manual reporting steps.
- Analysed material, labour, and overhead costs across 50+ projects — improved cost accuracy by 20% and identified Rs.10+ lakhs in annual savings.
- Administered PostgreSQL on production Linux VM — schema design, query optimisation, and data integrity for manufacturing analytics.

Data Analyst Intern · Essart India Pvt. Ltd. · Delhi NCR

Sept 2024 – Dec 2024

- Optimised SQL queries (JOINS, window functions, subqueries) to extract and analyse grievance data.
- Designed interactive Power BI dashboards for grievance trends, resolution times, and department performance.
- Used Python (Pandas) for data cleaning, transformation, and analysis to uncover key business insights.
- Re-engineered service processes across 10 departments — restructured communication flow and cut paperwork by 75%.

Software Development Intern · Spring Boot / Java · VVDN Technologies · Gurugram

Mar 2024 – Jul 2024

- Built RESTful APIs using Spring Boot (Java) following MVC architecture and REST best practices.
- Developed an Email Scheduler module — cron-based scheduling, SMTP integration, and dynamic template rendering.
- Implemented full CRUD operations for user management and task tracking using Spring Data JPA + MySQL.
- Conducted API testing in Postman and documented API contracts.

PROJECTS

Marketing Analysis for E-Commerce · Python · Pandas · Seaborn · RFM Feb 2025

- Performed RFM customer segmentation to enable targeted marketing strategies.
- Evaluated campaign KPIs to optimise marketing spend and support data-driven decisions.

Credit Consumption Prediction — Leading Bank · Python · ML · EDA May 2025

- Built ML regression models to predict credit card consumption over 3 months using 40+ engineered features.
- Validated models (Linear Regression, Decision Trees) optimised on RMSPE metric.

EDUCATION

Bachelor of Computer Science | GPA: 7.2 · Maharshi Dayanand University · Rohtak

Jun 2020 – Aug 2024

Coursework: DBMS, Java, Python, Machine Learning

CERTIFICATIONS

- **Advanced Data Science Bootcamp** · Analytix Labs, Gurgaon · May 2025
- **Zomato Data Analysis Using Python** · WSCUBE Tech · Oct 2024
- **GCP Professional Data Engineer** · In Progress · 2025–2026